Name: _____

Class: _____

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Algorithms 4 — Sorting: Bubble, Insertion and Merge — ANSWERS

Last lesson ended on one rule: **binary search needs a sorted list**. Putting a list in order is a job of its own, and OCR names three standard ways to do it. In the exam you **apply** each one to a list of data and show every stage, so this sheet is all stepping and tracing.

OCR J277 specification, 2.1.3 Searching and sorting algorithms

Required

- ✓ Understand the main steps of the algorithm and the segments of code in it
- ✓ Understand any pre-requisites of an algorithm
- ✓ Apply the algorithm to a data set
- ✓ Identify an algorithm if given the code, pseudocode or Exam Reference Language for it

Not required

- × To remember the code, pseudocode or Exam Reference Language for these algorithms

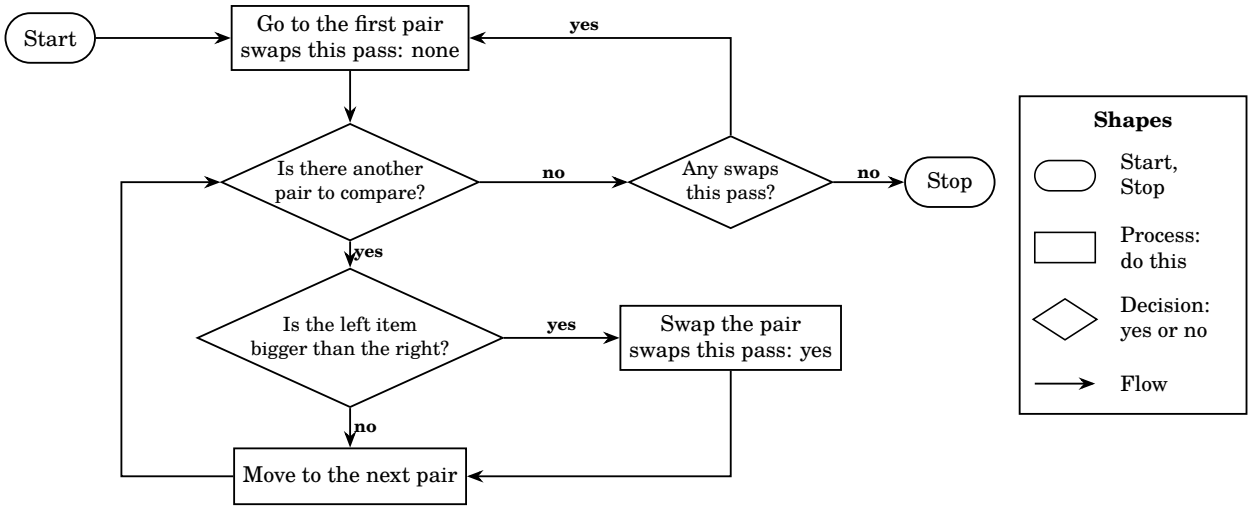
1 Sort it by hand first

Write each of these numbers on its own scrap of paper, mix them up, and put them in ascending order on the desk:

7 3 9 2 6 4

Now try • Finish the sentence that describes **your method**: "I started by … and then I …"

The same job as a flowchart. Every box is something your hands did. A **pass** is one trip along the list; the algorithm repeats passes until one pass needs no swaps.



2 Bubble sort

VOCAB **Bubble sort** (OCR 2.1.3) — compares each pair of neighbouring items from the start of the list and swaps the pair when it is in the wrong order, then repeats the whole pass until a pass makes no swaps

Watch it run: `rawsonvault.com/cs/gcse/2-1-algorithms/sorting-animations/bubble-sort.html`

The steps.

1. **Compare**: Start at the first item. Compare it with its neighbour on the right.
2. **Swap**: If the left item is **bigger**, swap the two. If not, leave them.
3. **Finish the pass**: Carry on to the end of the list: that is one **pass**.
4. **Count the swaps**: Count the swaps in the pass. If there was at least one, start another pass from the left. If there were **none**, the list is sorted: stop.

Worked example, pass 1. The list is `7 3 9 2 6 4`. Five pairs are compared. At each comparison the pair is outlined; the row below shows the list after it.

Compare 1	**7** **3** 9 2 6 4	7 is bigger than 3: **swap**
Compare 2	3 **7** **9** 2 6 4	7 is not bigger than 9: **no swap**
Compare 3	3 7 **9** **2** 6 4	9 is bigger than 2: **swap**
Compare 4	3 7 2 **9** **6** 4	9 is bigger than 6: **swap**
Compare 5	3 7 2 6 **9** **4**	9 is bigger than 4: **swap**
End of pass 1	3 7 2 6 4 **9**	4 swaps: another pass; 9 is in place

Pass 1 made 4 swaps, so another pass is needed. The biggest item, 9, has moved right until it reached the end: after every pass the next biggest item is in its final place.

How a swap works. Two boxes cannot swap contents directly, so a program uses a third box, `temp`: copy the left item into `temp`, copy the right item over the left, then copy `temp` into the right.

Now try • Complete the pass table. Each row is one whole pass, starting from the list at the end of the pass before. Stop on the pass that makes no swaps.

Pass	List at the end of the pass	Swaps	Another pass?
1	3 7 2 6 4 9	4	yes, there were swaps
2	3 2 6 4 7 9	3	yes
3	2 3 4 6 7 9	2	yes
4	2 3 4 6 7 9	0	no: stop, the list is sorted

The same algorithm as code. The exam shows it in OCR ERL; you can type the Python beside it and run it. The `swapped` flag is what records whether a pass made any swap.

OCR ERL (what the exam shows)

```
1  swapped = true
2  while swapped
3    swapped = false
4    for i = 0 to numbers.length - 2
5      if numbers[i] > numbers[i + 1] then
6        temp = numbers[i]
7        numbers[i] = numbers[i + 1]
8        numbers[i + 1] = temp
9        swapped = true
10     endif
11   next i
12 endwhile
```

PYTHON (what you type)

```
1  def bubble_sort(items):
2      swapped = True
3      while swapped:
4          swapped = False
5          for i in range(len(items) - 1):
6              if items[i] > items[i + 1]:
7                  temp = items[i]
8                  items[i] = items[i + 1]
9                  items[i + 1] = temp
10                 swapped = True
11         print(items)
12
13 bubble_sort([7, 3, 9, 2, 6, 4])
```

Now try • Type the Python and run it. Then match each job to its ERL line number.

The line that …	ERL line
compares a pair of neighbours	5
copies the left item into the spare box	6
records that this pass made a swap	9
keeps the passes going while the last pass had a swap	2

The questions you will meet on bubble sort. Every bubble sort question in the open past-paper pool asks for one of these five things.

The question asks you to	What to write	Marks
Show the stages of a bubble sort on a list	one row per pass (or per swap), writing the whole list each time	4
Explain the purpose of the Boolean variable	it records whether a swap has taken place in the pass; the `while` line checks it to decide whether to go round again	2
Explain the lines that swap two items	they swap `items[p]` and `items[p + 1]` when the left one is bigger, using `temp` so that neither value is lost	2
Name two other sorting algorithms	insertion sort and merge sort	2
Give an advantage of merge sort over bubble sort	it is faster on long lists, and its running time does not depend on how ordered the list was	2

Worked example. This one is written for this sheet, in the style of the exam question that follows it. The answer is shown, so you can see what a full answer looks like.

Show the stages of a bubble sort when applied to this list of words, A to Z.

`mango` `apple` `zebra` `lemon` `grape` `kiwi`

Method. Write the starting list. Take one pass from left to right, swapping each pair where the left word comes later in the alphabet. Write the *whole* list at the end of the pass. Go round again, and stop after a pass that makes no swaps.

Stage	**The whole list**						**Swaps**
start	`mango`	`apple`	`zebra`	`lemon`	`grape`	`kiwi`	
pass 1	`apple`	`mango`	`lemon`	`grape`	`kiwi`	`zebra`	4
pass 2	`apple`	`lemon`	`grape`	`kiwi`	`mango`	`zebra`	3
pass 3	`apple`	`grape`	`kiwi`	`lemon`	`mango`	`zebra`	2
pass 4	`apple`	`grape`	`kiwi`	`lemon`	`mango`	`zebra`	0

Pass 4 makes no swaps, so the list is sorted and the answer ends there. After pass 1 the last word, zebra, is already in its final place; after pass 2, mango is too. In pass 1, `mango` is compared with `apple` and swaps; then with `zebra`, and stays; `zebra` then swaps past `lemon`, `grape` and `kiwi`.

The real exam questions for this section are on the printed copy used in class.

3 Insertion sort

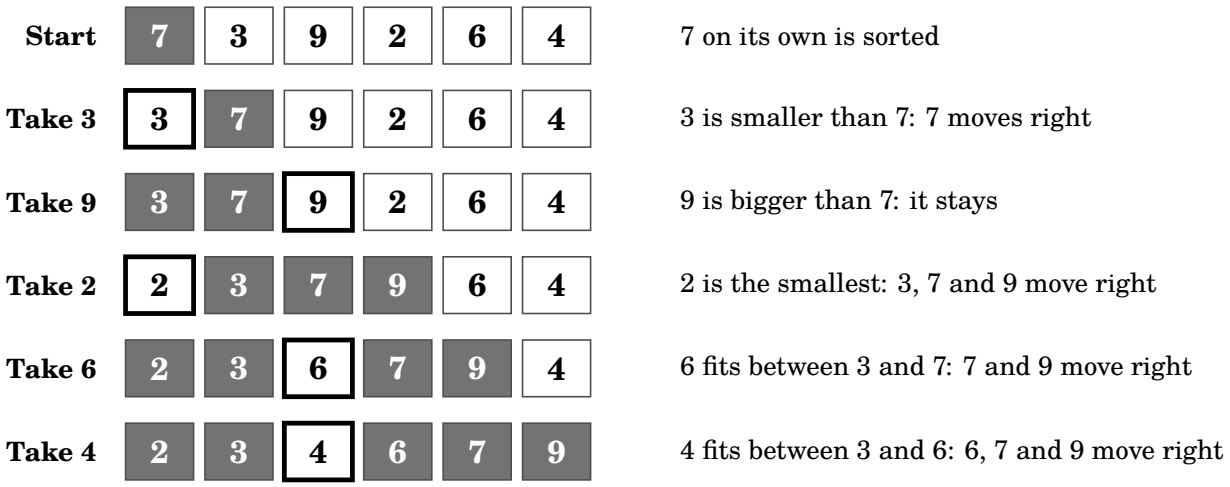
VOCAB **Insertion sort** (OCR 2.1.3) — splits the list into a sorted part on the left and an unsorted part on the right, then takes the unsorted items one at a time and inserts each into its correct place in the sorted part

Watch it run: `rawsonvault.com/cs/gcse/2-1-algorithms/sorting-animations/insertion-sort.html`

The steps.

1. **Start**: The **first item** on its own is the sorted part. Everything else is unsorted.
2. **Compare**: Take the first unsorted item. Compare it with the sorted items, working leftwards from the biggest, until you find where it belongs.
3. **Insert**: Insert it there. The bigger sorted items each move one place right.
4. **Repeat**: The sorted part is now one item longer. Repeat until the unsorted part is empty.

Worked example. The same list, `7 3 9 2 6 4`. The dark cards are the **sorted part**. The white card with the heavy outline is the item just inserted.



The same algorithm as code. The strips above are this program running on `7 3 9 2 6 4`. Type the Python and run it: it prints the list after each card is put down, the same five lists as the strip.

PYTHON (what you type)

```
def insertion_sort(items):
    for index in range(1, len(items)):
        item_to_insert = items[index]
        position = index - 1
        while position >= 0 and items[position] > item_to_insert:
            items[position + 1] = items[position]
            position = position - 1
        items[position + 1] = item_to_insert
        print(items)

insertion_sort([7, 3, 9, 2, 6, 4])
```

OCR ERL (what the exam shows)

```
1 for index = 1 to items.length - 1
2   itemToInsert = items[index]
3   position = index - 1
4   while position >= 0 AND items[position] > itemToInsert
5     items[position + 1] = items[position]
6     position = position - 1
7   endwhile
8   items[position + 1] = itemToInsert
9 next index
```

Python line	What it does on the cards
2	takes the next card from the unsorted part; `index` is its place
3–4	holds that card in hand, and points at the sorted card just to its left
5	asks whether that sorted card is bigger than the card in hand (and that a sorted card is still there)
6–7	slides the bigger sorted card one place right, then points at the next sorted card to the left
8	puts the card in hand down in the gap that is left

Worked example 2: seven cards. The same steps on seven cards. Dark cards are the sorted part; the heavy outline is the card just inserted.

Start	9	3	10	2	8	5	1	9 on its own is sorted
Take 3	3	9	10	2	8	5	1	3 is the smallest so far: 1 item moves right
Take 10	3	9	10	2	8	5	1	10 is bigger than 9: it stays
Take 2	2	3	9	10	8	5	1	2 is the smallest so far: 3 items move right
Take 8	2	3	8	9	10	5	1	8 fits after 3: 2 items move right
Take 5	2	3	5	8	9	10	1	5 fits after 3: 3 items move right
Take 1	1	2	3	5	8	9	10	1 is the smallest so far: 6 items move right

Worked example 3: medals. The number of medals won by five countries at the 1998 Winter Olympics.

Start	15	25	13	29	18	15 on its own is sorted
Take 25	15	25	13	29	18	25 is bigger than 15: it stays
Take 13	13	15	25	29	18	13 is the smallest so far: 2 items move right
Take 29	13	15	25	29	18	29 is bigger than 25: it stays
Take 18	13	15	18	25	29	18 fits after 15: 2 items move right

Worked example 4: words. Words are put in alphabetical order, A to Z.

Start	**Sydney**	**Athens**	**Turin**	**Beijing**	**London**	Sydney on its own is sorted
Step 1	**Athens**	**Sydney**	**Turin**	**Beijing**	**London**	Athens goes first: 1 item moves right
Step 2	**Athens**	**Sydney**	**Turin**	**Beijing**	**London**	Turin comes after Sydney: it stays
Step 3	**Athens**	**Beijing**	**Sydney**	**Turin**	**London**	Beijing goes after Athens: 2 items move right
Step 4	**Athens**	**Beijing**	**London**	**Sydney**	**Turin**	London goes after Beijing: 2 items move right

Examples 2 to 4 use the data from the insertion sort page of Isaac Computer Science.

Now try • Insertion sort a new list, `5 1 4 8 2 7`. Take the items in turn and write the whole list on each row, with the bar in its new place.

Item taken	Sorted part | Unsorted part	What happened
(start)	5 | 1 4 8 2 7	5 on its own is sorted
1	1 5 | 4 8 2 7	1 is the smallest: 5 moves right
4	1 4 5 | 8 2 7	4 fits between 1 and 5: 5 moves right
8	1 4 5 8 | 2 7	8 is bigger than 5: it stays
2	1 2 4 5 8 | 7	2 fits between 1 and 4: 4, 5 and 8 move right
7	1 2 4 5 7 8 |	7 fits between 5 and 8: 8 moves right

Write it: insertion sort in Python. The outer `for` line starts at index 1, because the item at index 0 is the sorted part to begin with. `index` is the item to insert, and `position` walks left through the sorted part. The `while` line tests `position >= 0` first, so `items[position]` is only read when `position` names a real item.

Now try • Task 1. Fill each gap so that the function sorts the list.

```
def insertion_sort(items):
    for index in range(1, len(items)):
        item_to_insert = items[index]
        position = index - 1
        while position >= 0 and items[position] > item_to_insert:
            items[position + 1] = items[position]
            position = position - 1
        items[position + 1] = item_to_insert
        print(items)
```

Now try • Task 2. Run `insertion_sort([6, 2, 8, 3, 1])` in your head. Write the list that `print(items)` shows after each item is inserted, and the number of items that were copied up one place each time.

Item inserted	Output of `print(items)`	Items copied up
2	`[2, 6, 8, 3, 1]`	1
8	`[2, 6, 8, 3, 1]`	0
3	`[2, 3, 6, 8, 1]`	2
1	`[1, 2, 3, 6, 8]`	4

Now try • Task 3 (stretch). The list is already in ascending order, `[1, 2, 3, 4, 5]`. How many items does the insertion sort copy up in total, and how many times does the `while` loop body run?

ANSWER

None. Each item to insert is bigger than the sorted item before it, so the `while` condition is false at once and the body runs 0 times. Each item is written back into the place it came from.

4 Merge sort

VOCAB **Merge sort** (OCR 2.1.3) — a divide and conquer algorithm: it splits the list in halves until every list holds one item, then merges the lists back together in pairs, putting the items in order as it merges

Watch it run: `rawsonvault.com/cs/gcse/2-1-algorithms/sorting-animations/merge-sort.html`

The steps.

1. **Divide**: Split the list into single items. A list of one item is already sorted.
2. **Merge**: Take two neighbouring lists. Compare their **first** items, move the smaller into the new list, and repeat until one list is empty. Add whatever is left of the other list on the end.
3. **Keep merging**: Repeat step 2 with the new lists until a single list remains.

Worked example, one merge. Merge `3 7` with `2 9`.

Step	Left list	Right list	Smaller first item	New list
1	3 7	2 9	2	2
2	3 7	9	3	2 3
3	7	9	7	2 3 7
4	(empty)	9	9 is left over	2 3 7 9

Worked example, the whole sort. Eight items, divided into single items, then merged back in three rounds.

Lists of 1	7	3	9	2	6	4	8	1
Lists of 2	3 7		2 9		4 6		1 8	
Lists of 4	2 3 7 9				1 4 6 8			
Lists of 8	1 2 3 4 6 7 8 9							

Each merge produces a list that is already in order, so no list is ever sorted again afterwards: the merging *is* the sorting.

Now try • Merge sort this list. The row of single items is done for you. Fill in the three merge rounds, drawing each new list across the boxes it covers.

Lists of 1	5	1	8	3	7	2	6	4
Lists of 2	1 5		3 8		2 7		4 6	
Lists of 4	1 3 5 8				2 4 6 7			
Lists of 8	1 2 3 4 5 6 7 8							

The same algorithm as code. Merge sort in Python is two functions. `merge` joins two sorted lists into one. `merge_sort` splits the list in half, calls itself on each half, and merges the two results. Type it and run it on the list from your worked example: the exam asks you to follow merge sort on a list of data, and the program shows every merge it makes.

PYTHON (what you type)

```
1  def merge(left, right):
2      merged = []
3      i = 0
4      j = 0
5      while i < len(left) and j < len(right):
6          if left[i] <= right[j]:
7              merged.append(left[i])
8              i = i + 1
9          else:
10             merged.append(right[j])
11             j = j + 1
12     merged = merged + left[i:] + right[j:]
13     return merged
14
15 def merge_sort(items):
16     if len(items) <= 1:
17         return items
18     middle = (len(items) + 1) // 2
19     left = merge_sort(items[:middle])
20     right = merge_sort(items[middle:])
21     merged = merge(left, right)
22     print(left, right, "->", merged)
23     return merged
24
25 merge_sort([7, 3, 9, 2, 6, 4, 8, 1])
```

What it prints. One line for each merge, in the order the program does them.

```
[7] [3] -> [3, 7]
[9] [2] -> [2, 9]
[3, 7] [2, 9] -> [2, 3, 7, 9]
[6] [4] -> [4, 6]
[8] [1] -> [1, 8]
[4, 6] [1, 8] -> [1, 4, 6, 8]
[2, 3, 7, 9] [1, 4, 6, 8] -> [1, 2, 3, 4, 6, 7, 8, 9]
```

Python line	What it does
1–13	`merge`: the merge step from the table above, as a function
16–17	a list of one item, or none, is already sorted, so it is handed back as it is
18	finds the middle; with an odd number of items the first half takes the extra one
19–20	sorts each half by calling `merge_sort` on it
21	merges the two sorted halves
22	prints the merge, so the lists of 2, then 4, then 8 can be read off

Write it: the merge step in Python. The merge is the part of merge sort that does the sorting, and it is short. `i` marks the first unused item of `left` and `j` the first unused item of `right`. Each trip round the loop moves the smaller of those two items into `merged`. When one list runs out, the rest of the other list goes on the end, already in order.

Now try • Task 1. Fill each gap so that `merge` joins two sorted lists into one sorted list.

```
def merge(left, right):
    merged = []
    i = 0
    j = 0
    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            merged.append(left[i])
            i = i + 1
        else:
            merged.append(right[j])
            j = j + 1
    merged = merged + left[i:] + right[j:]
    return merged
```

Now try • Task 2. Run `merge([2, 5, 9], [3, 4])` in your head. Write `merged` after each comparison, then after the leftover items are added.

Step	`merged`
Compare 2 and 3	`[2]`
Compare 5 and 3	`[2, 3]`
Compare 5 and 4	`[2, 3, 4]`
Add the leftover items	`[2, 3, 4, 5, 9]`

Now try • Task 3 (stretch). A merge sort splits `[15, 25, 13, 29, 18]` in half. With an odd number of items the first half takes the extra one. Write the two halves.

ANSWER

`[15, 25, 13]` and `[29, 18]`. Five items split 3 and 2.

5 Three sorts side by side

	Bubble sort	Insertion sort	Merge sort
The idea	swap neighbours that are the wrong way round, pass after pass	grow a sorted part on the left, one inserted item at a time	divide into single items, then merge in order
What stops it	a pass with no swaps	the unsorted part is empty	one list is left
The list at the start	any order	any order	any order
On a long list	slowest of the three	faster than bubble sort	fastest of the three
Comparisons on `7 3 9 2 6 4`	20	12	10
Why choose it	the simplest idea, with the shortest code to follow	fewer comparisons than bubble sort, and each card is put in its place once	the fastest on long lists, and the same speed however the list starts

The comparison counts come from running each sort on that list: bubble sort checks all 5 pairs on all 4 passes, insertion sort stops looking as soon as a card finds its place, and merge sort compares the front items of two lists. In the exam, the advantage of insertion sort over bubble sort is that it is more efficient: fewer comparisons, more so when the data is more scrambled.

Now try • Name the sort each statement describes: **bubble**, **insertion** or **merge**.

	Statement	Sort
a	The list is split into single items before anything is compared.	merge
b	Each pass compares neighbouring pairs from the left of the list.	bubble
c	The first item is treated as a sorted list of one.	insertion
d	It stops when a pass completes without a swap.	bubble
e	Each new item is slotted into its place among the items already in order.	insertion
f	Two ordered lists are combined by repeatedly taking the smaller first item.	merge

Sort first, then search. A sorted list is what binary search was waiting for. Sort `7 3 9 2 6 4` with any of the three sorts, and `2 3 4 6 7 9` is ready for it.

Now try • Use binary search on the sorted list `2 3 4 6 7 9` to find **9**. List each middle item you check.

ANSWER

Checks: 4 (index 2), 7 (index 4), 9 (index 5). Start 0, end 5 gives middle 2. 4 < 9, so start = 3. (3 + 5) DIV 2 = 4: item 7. 7 < 9, so start = 5. (5 + 5) DIV 2 = 5: item 9, found.

The exam answers, one line each

Asked for	Write
Show the stages of a bubble sort	One row per pass (or per swap), writing the whole list each time, the biggest remaining item moving to the right.
Show the stages of an insertion sort	One row per inserted item, the sorted part growing by one each row.
Show the stages of a merge sort	Split into single items, then merge into lists of 2, then 4, then the whole list, each list in order.
An advantage of merge sort	It is faster on long lists, and its running time does not depend on how ordered the list was to begin with.